Mapping Political Proposals in 8 Dimensions: Reliability, Calibration Bias, and What AI Models Cannot Agree On

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Abstract

AI tools for political sensemaking are proliferating, yet the methodology behind most of them remains opaque. Politikea is a civic platform that maps users across two ideological axes based on their votes on policy proposals. This paper stress-tests that 2-axis framework by scoring N=2,977 Spanish proposals across 8 independent ideological axes using large language models. Within-model reliability is strong: ICC(2,1) $\geq$ 0.88 across all axes, with 82% of proposals passing reliability thresholds. Cross-model triangulation against Gemini 2.5 Flash and llama3.1:8b reveals structurally distinct calibration biases — systematic positive offset vs. task-level overload — neither passing the publication gate. Dimensionality analysis shows the Politikea product pair ranks 11th of 36 possible 1- and 2-axis projections (R^2 = 0.595), while the best single-axis projection ranks 23rd (R^2 = 0.456), quantifying what unidimensional models lose. We release a full scoring toolkit for replication and extension.

Keywords: political ideology, large language models, text annotation, reliability, calibration, computational social science

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AI tools for political sensemaking are being built right now, by governments, platforms, and civic organizations. Most of the methodology behind them is invisible. We think that needs to change, not because any single team has the right answer, but because the field moves faster and more responsibly when the work is in the open. Politikea is a civic AI platform for understanding political disagreement across multiple ideological dimensions. This document shares what we built, what we found, what we got wrong, and what we still do not know. We are releasing the toolkit so others can run the same analysis on their own data and tell us where we are wrong. The more people stress-test this, the better it gets for everyone.

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1 Introduction

Political participation platforms face a fundamental design problem: how do you represent where someone stands politically without reducing them to a point on a line? The traditional

left-right spectrum is cognitively convenient but analytically impoverished. It collapses economically redistributive positions with culturally progressive ones, conflates support for decentralization with libertarianism, and treats technocratic governance preferences as orthogonal noise. A citizen who favors strong public healthcare but opposes immigration quotas does not fit cleanly on a left-right axis. Neither does a citizen who wants a decentralized state but supports active redistribution.

Politikea takes a different approach. Rather than asking users to place themselves on a spectrum, it asks them to vote on concrete policy proposals — *politikas* — and maps each user's position across two ideological axes: equity vs. freedom (`aequitas_libertas`) and authority vs. decentralisation (`imperium_anarkhia`). The user's position on this 2D map emerges from their vote history. This document reports the methodology and Phase 1 findings from a broader 8-axis scoring exercise on N=2,977 Spanish political proposals — an internal research effort to uncover biases in our 2-axis framework, stress-test its information retention, and push the methodology further for others to build on.

1.1 The 8-Axis Framework

Each proposal is scored on a continuous −100 to +100 scale across 8 axes. A score of 0 means balanced or ambiguous; ±20 is a weak preference; ±50 is strong; ±80 to ±100 is dominant. The axes are:

Table 1: The 8-axis ideological framework. All axes scored −100 to +100. Score of 0 = balanced or ambiguous; ±20 = weak preference; ±50 = strong; ±80–±100 = dominant orientation. Axes are independent of each other.

Axis	Negative pole (−100)	Positive pole (+100)
aequitas_libertas (equity vs. freedom)	Full redistribution, equality of outcome	Individual freedom, minimal redistribution
imperium_anarkhia (authority vs. decentralization)	Centralized state authority	Decentralization, self-organization
universalism_particularism (universal rules vs. group-specific)	Universal rules and rights for all	Group-specific or context-specific policies
market_collective_allocation (market vs. collective)	Market allocation of resources	Collective or democratic allocation
inequality_acceptance_correction (natural vs. structural)	Inequality is acceptable and natural	Inequality is structural and must be corrected
individual_socialized_risk (individual vs. pooled risk)	Individuals absorb their own risk	Risk pooled collectively
progressivism_preservation (change vs. tradition)	Active reform, embrace of change	Stability, tradition, continuity
technocracy_populism (experts vs. people)	Expert-led governance	Popular-will primacy

The full scoring prompt is reproduced verbatim in Appendix A. The prompt is the operational specification - understanding what any score means requires reading it.

1.2 Running Example

Throughout this paper we use a specific proposal to anchor the mathematical concepts in something concrete: *"Expropiación de viviendas vacías — proponer que las viviendas vacías propiedad de entidades bancarias o fondos de inversión sean expropiadas y destinadas a alquiler social."* (Expropriation of empty housing — proposing that empty properties owned by banking entities or investment funds be expropriated and allocated to social housing.) This proposal appears in our triangulation worst-case analysis, making it doubly useful: it illustrates both what the model found and where cross-model disagreement was sharpest.

1.3 Scope and Limitations Stated Upfront

- N=2,977 proposals; Spanish language; single platform (Politikea)
- Labels from GPT-5.2 only; no human annotator ground truth in Phase 1
- Cross-model triangulation attempted with Gemini 2.5 Flash and llama3.1:8b; both fail the agreement gate (this is a finding, discussed in Section 4.2)
- All results are within-model reliability claims, not ground-truth-validated

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2 Related Work

2.1 Political Compass Frameworks

The Political Compass (Pace and Marshall 2001) was among the first widely adopted attempts to extend political self-placement beyond a single axis, adding an authoritarian-libertarian dimension orthogonal to the economic left-right axis. It exposed how the traditional spectrum collapsed distinct ideological dimensions. However, it relies on self-report, is sensitive to question framing, and has been criticized for ideological biases in the item selection. Our approach sidesteps self-report entirely: instead of asking users to answer questions about themselves, it scores the concrete policy proposals they engage with using AI, replacing subjective questionnaire items with empirically validated text annotations.

2.2 Computational Ideology Scaling

Wordfish (Slapin and Proksch 2008) infers ideological positions from word frequencies in political texts using a Bayesian scaling model. It is powerful for party manifestos and speeches but assumes a single latent dimension and requires sufficient text length for reliable estimation. LIWC (Pennebaker et al. 2001) and its successors offer rich psycholinguistic dictionaries that capture emotional and cognitive dimensions of language but do not map directly to political ideology. Embedding-based methods (e.g., Rodriguez and Spirling 2022) project political texts into dense semantic spaces and measure positional similarity, but they too tend to collapse the ideological space into a small number of principal directions without explicit axis semantics. Our approach is complementary: we use AI to score proposals on pre-defined axes with explicit definitions, then validate those scores empirically.

2.3 LLM Annotation Reliability

Gilardi et al. (2023) showed that GPT-3.5 outperformed crowdworkers on several text annotation tasks, opening the door to large-scale LLM-based labeling for social science research. Møller et al. (2023) studied the reliability of LLM annotations across multiple models and tasks, finding that agreement varies substantially by task type and that LLMs can exhibit systematic biases that differ from human annotator biases. Our work extends this line of research to a specific high-stakes domain — continuous ideological scoring — and introduces a cross-model triangulation protocol as a diagnostic tool. We find that model-specific calibration biases are detectable, characterizable, and structurally distinct across models.

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3 Methodology

3.1 Labeling Pipeline

Each proposal was scored by GPT-5.2 using a structured output prompt (Appendix A) that specifies all 8 axes, their score anchors, and the JSON output schema. The model was called N=13 times per proposal on average, with slight variations in sampling parameters to introduce stochasticity. This multi-run design is the foundation of our reliability claims: we treat consistency across runs as evidence of signal, not noise.

Structured output: the prompt enforces a strict JSON schema with per-axis scores (float, -100 to $+100$) and 0–4 rationale spans with character offsets into the proposal text. The schema prevents hallucination of extra keys and forces the model to reference the proposal text directly.

Why multi-run: a single LLM call on a political proposal is susceptible to positional effects, sampling noise, and prompt-order artifacts. Running 13 times and aggregating via mean and sign agreement gives us a stable estimate and a per-proposal uncertainty measure.

3.2 Reliability Filtering

After scoring, each proposal is filtered for validity. A proposal is considered *valid* if:

- At least 10 runs completed
- Global confidence ≥ 0.5 (mean across axes)
- Per-axis confidence ≥ 0.4
- At least 6 of 8 axes are stable (sign agreement ≥ 0.6 on that axis)

This yields 2,442 valid proposals (82.0%) with bootstrapped CI [80.6%, 83.4%]. Results are robust across threshold variations; the highest valid rate achievable with looser thresholds is 87.9%.

3.3 Cross-Model Triangulation Protocol

To audit the GPT-5.2 labels, we drew a stratified sample of 399 proposals and re-scored them using two independent models: Gemini 2.5 Flash (frontier model, free-tier API) and

llama3.1:8b (open-weight model, self-hosted). Model selection was budget-constrained: GPT-5.2 scoring of the full corpus cost approximately $200; the audit layer was designed to run at near-zero marginal cost. This is a deliberate trade-off, not a methodological ideal; stronger audit models (GPT-4o, Claude 3.5, larger llama variants) would be preferable and are a stated priority for Phase 2. We report the results here because the failure patterns are informative regardless: both models fail the gate, but for structurally different reasons that expose distinct model-specific failure modes. Each audit model was called 3 times per proposal at temperature 0.2, and scores were aggregated by median. We then computed directional agreement (fraction of axes where audit and GPT signs match) and mean absolute error (MAE) against a publication gate of directional agreement ≥ 0.85 and MAE ≤ 15. The gate thresholds are a design choice, not an empirically calibrated baseline; human inter-annotator agreement on continuous ideological scales has not been measured for this task and is scoped for Phase 2. The gate functions as a diagnostic tool, not a binary quality verdict.

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4 Results

4.1 Within-Model Reliability: The Foundation Holds

Before anything else can be claimed, we need to know whether GPT-5.2 applies the axes consistently. The answer is yes, strongly (Table 2).

Table 2: Within-model reliability. ICC(2,1) = intraclass correlation coefficient, two-way mixed model, single measures. Values ≥ 0.75 are considered excellent. Mean Sign Agreement = fraction of runs where the axis score has the same sign as the mean. Based on N=13 runs per proposal (GPT-5.2), N=2,977 proposals total.

Axis	ICC(2,1)	Mean Sign Agr.
aequitas_libertas (equity vs. freedom)	0.967	0.963
imperium_anarkhia (authority vs. decentralisation)	0.964	0.977
universalism_particularism (universal vs. group-specific)	0.885	0.936
market_collective_allocation (market vs. collective)	0.976	0.983
inequality_acceptance_correction (natural vs. structural)	0.969	0.981
individual_socialized_risk (individual vs. pooled)	0.958	0.973
progressivism_preservation (change vs. tradition)	0.937	0.969
technocracy_populism (experts vs. people)	0.931	0.949

ICC(2,1) values ≥ 0.88 across all axes indicate excellent inter-run agreement — the model applies the same interpretation consistently across repeated calls on the same text. In concrete terms: if we scored the housing expropriation proposal ten times, nine of those scores would land within 5 points of each other on every axis.

Why this matters: ICC is a within-model reliability claim, not a ground-truth validity claim. It tells us the labels are stable, not that they are correct. The cross-model audit (Section 4.2) addresses validity.

Overall: 2,442 of 2,977 proposals (82.0%) pass all reliability thresholds, with bootstrapped CI [80.6%, 83.4%]. The remaining 18% are excluded from downstream analyses. The current interpretation is that these are proposals the model itself cannot categorize consistently — possible signals of genuine ideological ambiguity in the text. However, we have not audited this tail: whether it clusters in specific policy categories, shares linguistic features, or instead reflects underspecified axes is an open question documented in the Limitations section.

4.2 Cross-Model Audit: AI Models Are Biased But Consistent

Both audit models fail the gate. But they fail for structurally different reasons, and that difference is the finding.

Gemini 2.5 Flash is *internally consistent* but *systematically offset*. Within-model axis SD = 4.27, instability rate 12.8% — the model gives nearly the same answer on the same proposal across its 3 runs. But its scores diverge from GPT-5.2 by an average MAE of 62.9 with directional agreement of only 0.396. Gemini scores positive across all 8 axes (mean scores +6.5 to +44.3), while GPT-5.2 shows both positive and negative means (−21.6 to +22.3) — see Figure 1. This is response-range bias, also known as acquiescence bias, where the model anchors toward the positive end of every axis regardless of content. The worst axis is `progressivism_preservation` (sign agreement 0.12, MAE = 89.96): Gemini consistently interprets the same proposals as tradition-preserving where GPT consistently interprets them as reform-oriented.

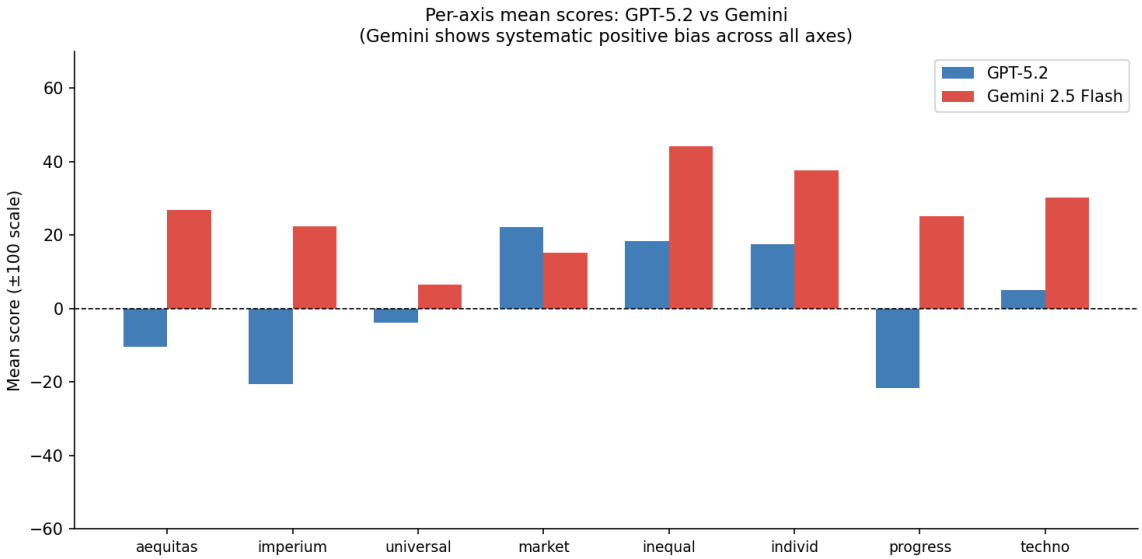

Figure 1: GPT-5.2 vs. Gemini 2.5 Flash per-axis mean scores. Gemini scores positive across all 8 axes — a response-range/acquiescence bias. GPT-5.2 tracks the actual ideological distribution of the corpus with both positive and negative means.

llama3.1:8b is *internally inconsistent* and *task-confused*. Within-model axis SD = 17.33, instability rate 62.0%. Round-number rate (scores as multiples of 10) = 70%, indicating the model treats the scoring task as a categorical choice rather than using the continuous scale. Per-axis mean offsets are large but mixed in direction, unlike Gemini's uniform positivity bias;

see Figure 2. This is task-overload combined with limited continuous-scale comprehension, not calibration divergence.

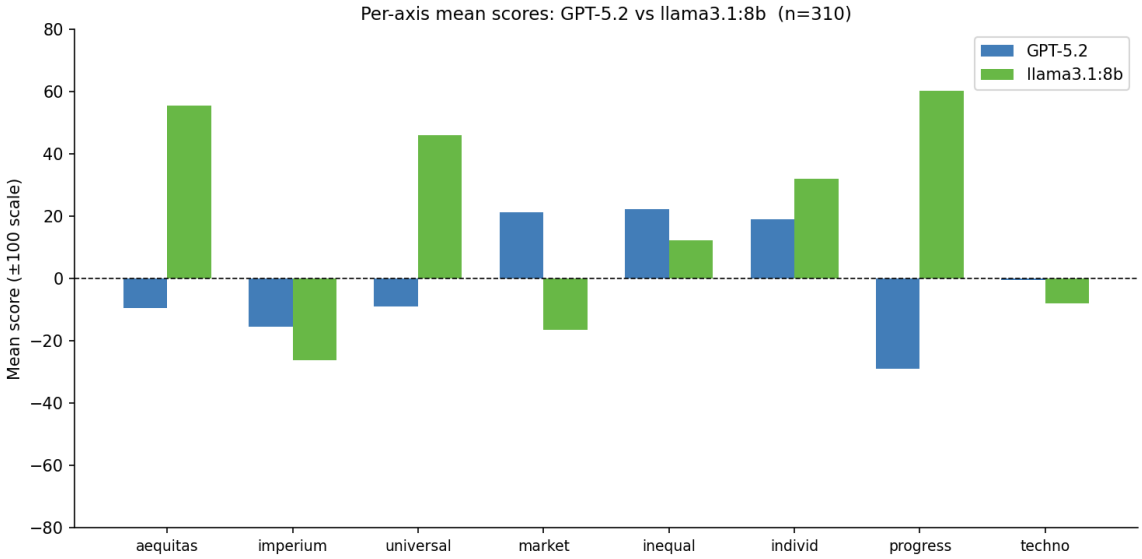

Figure 2: GPT-5.2 vs. llama3.1:8b per-axis mean scores. llama shows large mixed-direction offsets with axis-specific patterns, contrasting with Gemini's uniform positive shift.

The comparison is the finding: both models fail the gate, but for structurally different reasons. Gemini *understands the task* but applies it with a systematic offset that could be corrected via per-axis polarity anchors or offset normalization. Llama *misunderstands the task structure* and requires either prompt decomposition (one axis at a time, in Spanish — see Section 4.3) or a larger model.

What this comparison shows: all three models tested (GPT-5.2, Gemini 2.5 Flash, llama3.1:8b) produced internally consistent scores, yet each carried distinct calibration biases that became clearer when comparing outputs across models. Gemini's uniform positive shift would likely surface in routine inspection of score distributions; llama's quantization collapse is similarly visible in per-axis histograms. Cross-model comparison adds a systematic way to characterize these biases rather than relying on ad hoc checks, and is a priority for Phase 2 with stronger audit models.

4.2.1 Offset vs. Quantization: Two Distinct Failure Modes

Figure 3 places the two failure modes side by side at the proposal level. Gemini's scatter (top panel) shows a coherent diagonal structure; the model ranks proposals correctly relative to each other, but the entire cloud is shifted above the GPT-5.2 diagonal, most severely on `progressivism_preservation` where sign inversion is near-total. The failure is one of *calibration*, not comprehension. Llama's scatter (bottom panel) tells a different story: horizontal banding across all axes, where scores collapse to multiples of 10 regardless of the proposal content. The model is not miscalibrated; it is not using the scale at all. These are not two versions of the same problem; they require different fixes.

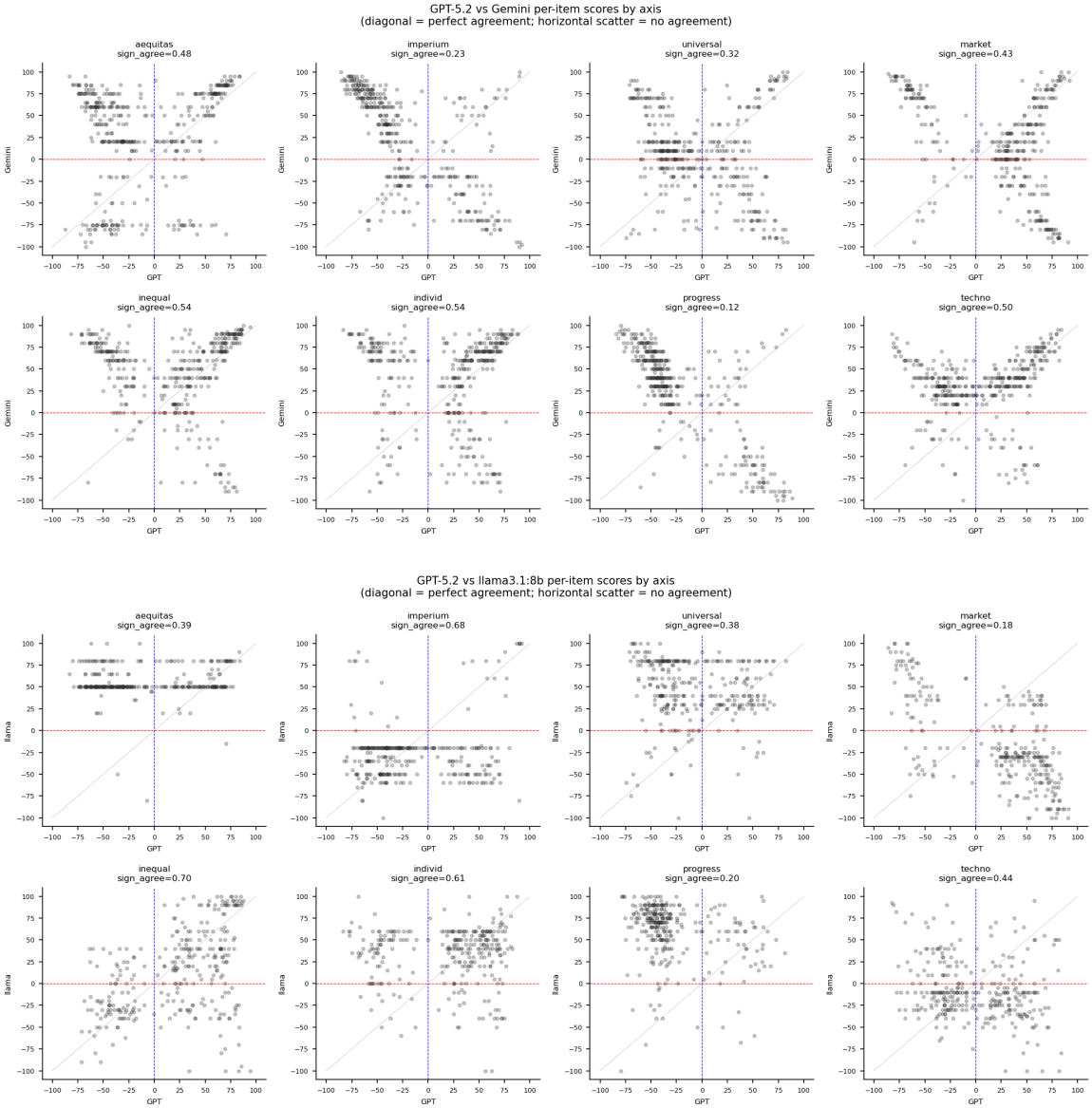

Figure 3: Per-item scatter plots comparing GPT-5.2 against both audit models. **Top**: Gemini 2.5 Flash — the cloud shifts above the diagonal on most axes (acquiescence bias), with `progressivism_preservation` showing near-total sign inversion. **Bottom**: llama3.1:8b — horizontal banding indicates score quantization, collapsing continuous scores to multiples of 10.

4.3 Prompt Ablation: Why llama Fails and How Far Prompt Engineering Gets

To understand llama's failure mode and explore recovery options, we tested 6 prompt variants on 50 proposals with 3 random seeds each (150 calls per variant, 900 total). **Clarification on Section 4.2 results**: the llama3.1:8b metrics reported in the cross-model audit (instability rate 62.0%, round-number rate 70%, GPT agreement 0.396) were obtained using the same prompt configuration applied to Gemini — this was intentional to keep the two audit models comparable under identical conditions. Section 4.3 then tests whether prompt engineering can close the gap for open-weight models. The full ablation results are in Table 3:

Table 3: Prompt ablation results for llama3.1:8b (N=50 proposals × 3 seeds = 150 calls/variant). Round-rate = fraction of scores that are exact multiples of 10. Stability = sign agreement across 3 runs. GPT agree = directional agreement vs. GPT-5.2 reference. Gate: GPT agree ≥ 85%. Latency on consumer hardware (Ollama, Apple Silicon). Tokens: median per item; Variant F totals reflect 8 per-axis calls. Descriptions in Section 7.8. Arrow indicates direction of improvement.

Variant	Round ↓	Stab. ↑	GPT agr. ↑	Lat. (s)	Tokens in/out
A: Minimal English baseline	99.7%	71.7%	36.2%	2.3	285 / 102
B: GPT-parity English	98.5%	66.1%	39.2%	3.2	370 / 102
C: Chain-of-thought (English)	71.2%	57.2%	59.2%	10.8	576 / 464
D: Nested reasoning JSON	2.1%	62.5%	61.8%	9.8	525 / 427
E: Few-shot calibration examples	9.7%	45.8%	47.5%	10.6	997 / 449
F: Spanish, one axis per call	7.1%	78.5%	65.2%	18.7	2,883 / 541

Stability vs. agreement measure different things. Within-model stability is precision (does the model give consistent answers across seeds?); GPT agreement is validity (do those answers match the reference labels?). These are independent — a variant can be stable and wrong (A: 71.7% stable, 36.2% agreement), or less stable but better aligned (C: 57.2% stable, 59.2% agreement). The simplest prompts (A, B) produce coarsely quantized but reproducible outputs; reasoning prompts (C, D) improve alignment but introduce variance as small differences in the chain-of-thought cascade into different final scores. Only Variant F breaks this trade-off: it is the sole **Pareto-optimal variant**, dominating all others on the stability–agreement frontier simultaneously.

Three findings stand out. First, few-shot examples *hurt* agreement — Variant E drops GPT agreement to 47.5% vs 36.2% for the baseline. The model pattern-matched on the ideological tone of the examples rather than using them as scale anchors; calibration-targeted few-shot design (with explicit boundary cases) remains an open problem. Second, nested reasoning (Variant D) eliminates quantization entirely and achieves 61.8% GPT agreement, but within-model stability remains modest at 62.5% — the reasoning process introduces variance across runs. Third, and most importantly for practitioners: **language matching and task decomposition matter more than prompt sophistication**. Variant F (Spanish prompt, one axis per call) achieves the highest within-model stability of any variant (78.5%) while maintaining the best GPT agreement (65.2%). The interpretation is that smaller models apply scoring tasks more consistently when operating in the same language as the text and when judging one dimension at a time — the cognitive overhead of cross-language inference and multi-axis simultaneous judgment are measurable sources of instability, not just stylistic concerns.

The cost of task decomposition. Variant F uses 8.8× the total tokens of baseline (3,424 vs 387 per item) because the item text is repeated in each of the 8 per-axis calls. On local Ollama this translates to wall-clock time only (18.7s vs 2.3s per item). On paid APIs the economics change: at GPT-4o rates ($2.50/M input, $10/M output), Variant F would cost ~$0.013/item ($12.62 per 1,000 items), making it the most expensive variant by far. This is a meaningful finding for practitioners: **the optimal prompt strategy for quality (F) is also the most expensive on paid APIs — self-hosting open-weight models eliminates this tension entirely**. For teams operating on paid APIs, Variant D offers the best agreement-per-token ratio at roughly half the token cost of F, albeit with lower stability.

Generalisation beyond Spanish: this finding should not be read as "use Spanish prompts." The correct generalisation is: *match the prompt language to the corpus language, and decompose multi-axis tasks into single-axis calls when using smaller open-weight models.* A model with limited multilingual capacity will produce more stable outputs when it does not need to simultaneously parse instructions in one language and text in another. This applies to any non-English civic corpus — Arabic, Portuguese, German, or otherwise. The choice of GPT-5.2 as primary annotator partially sidesteps this issue due to its stronger multilingual representations, but it remains a live concern for any replication using budget-constrained open-weight models.

Recommendation for open-source replication: use Variant F as the baseline prompt strategy — it is the only variant that achieves both high stability and competitive agreement. It still does not pass the publication gate (65.2% GPT agreement vs. 85% required); Variant F is the best available configuration for llama3.1:8b, not a validated annotator. A ≥30B model, or a small supervised fine-tune on GPT-labeled examples, would likely close the remaining gap. See Section 7.8 for full prompt descriptions for each variant.

4.4 What the 8D Space Looks Like on Real Proposals

With the reliability foundation established, we can examine what the 8D space reveals about the corpus. Table 8 shows the mean score per axis for the top 8 policy categories by proposal count; Figure 4 plots the top 7 ($N \geq 162$). The 4 remaining categories (Culture & Civic Education, Infrastructure & Housing, International Relations, Science/Energy/Environment) fall below the top-8 threshold and are omitted from both.

Two patterns stand out (Figure 4):

Security & Defence is the most striking outlier. It scores strongly pro-authority (imperium mean = −45.4), becoming the most statist category in the corpus. This is coherent: defence and security proposals tend to invoke centralised state power by nature. But it simultaneously scores near-neutral on redistribution (aequitas mean = −4.6) and slightly tradition-preserving (progressivism mean = +14.3, where +100 = tradition). This combination is politically interesting. It also scores the most particularist on universalism (+10.3 toward group-specific policies), consistent with security discourse that tends to frame policy around specific groups (nationals, citizens, threat actors) rather than universal principles. The interaction of strong authority-orientation with near-zero redistribution signal would be invisible on a left-right slider, which conflates the two.

Rights & Freedoms shows the opposite universalism pattern: the strongest universalist signal of any category in the visible 7 (universalism mean = −16.6, where −100 = universal

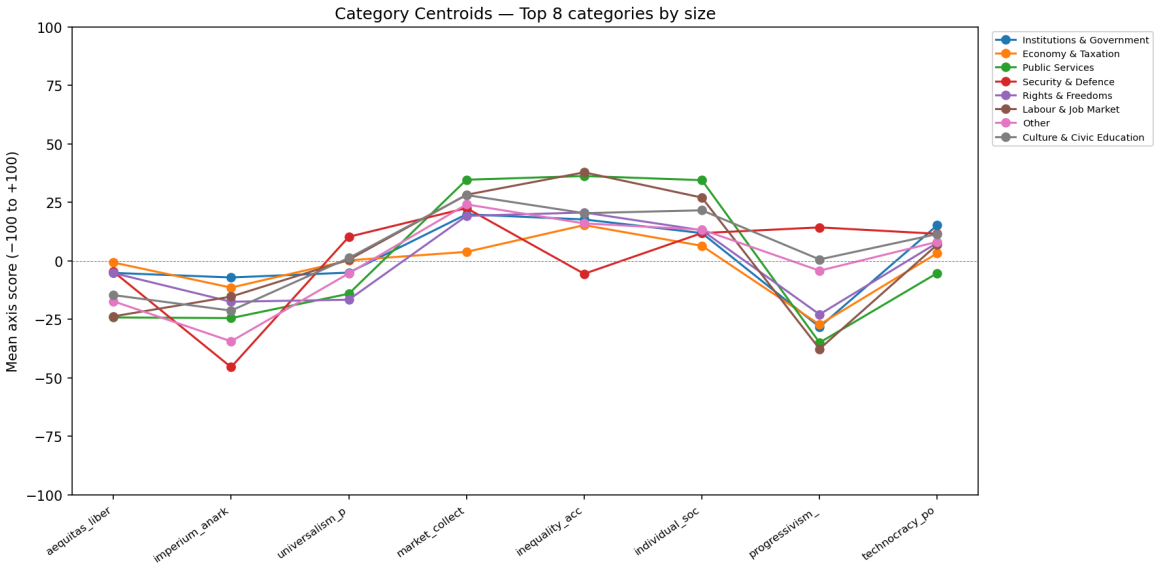

Figure 4: Category centroid profiles for the top 7 policy categories by proposal count (N $\geq$ 162). Each spoke represents one of the 8 ideological axes; the filled area shows the mean score for that category. Full scores for all 8 top categories, including Identity & Social Cohesion (N=80), are in Table 8.

rules for all), combined with reform orientation (progressivism = −22.9). This is noteworthy because rights discourse in contemporary politics is often associated with group-specific framing: rights for particular communities, identities, or minorities. The data shows the opposite. On the Politikea corpus, rights proposals seem to be framed in universal terms, applying to all citizens equally, and lean toward active reform rather than preservation of existing arrangements. The contrast with Security & Defence (particularist, statist, tradition-leaning) is sharp across three axes simultaneously.

Cross-category contradictions: the 8D framework opens the door to measuring whether ideological positions are consistent across policy areas in aggregate. Do voting patterns reflect the same ideological logic on labour market proposals as on public services? Phase 2 will explore this directly: do contradictions cluster by policy category, or do they vary by the order in which proposals are engaged with over time? We have the data structure to test this but want sufficient coverage before reporting a number.

4.5 What Text Alone Cannot Organize

A key question for any AI-based labeling system: does the model learn something the text does not already contain? We test this by measuring how well text similarity predicts 8D ideological similarity, and vice versa.

Global correlation: Spearman $\rho = 0.127$ (p = 0.005, 95% CI [0.040, 0.211]) between text similarity and 8D ideological similarity across 500 sampled nearest-neighbor pairs. This is a weak signal — real, but insufficient on its own.

Why the global correlation is weak: all Spanish political proposals share surface vocabulary (policy verbs, institutional names, topic nouns). Two housing proposals always look more similar to each other than to a pension proposal — regardless of ideology. When we restrict to within-category pairs (controlling for topic), the pooled ρ is 0.076 — both the global and within-category correlations are weak, indicating that text similarity and 8D

similarity measure largely independent signals. This independence is precisely why combining them works.

Neither signal alone is sufficient. Two concrete pairs from the corpus illustrate both failure modes (full worked examples with English glosses in Section 7.7):

Noise mode 1 — text similar, ideologically opposite: *"Prohibición de la educación concertada"* [Ban on state-subsidised private schools] vs *"Desregulación del sistema educativo"* [Deregulation of the education system] (text_sim = 0.793, 8D_sim = −0.708). Both proposals use nearly identical vocabulary — *education*, *schools* — but advocate opposite stances. A text-only deduplicator would wrongly merge them.

Noise mode 2 — ideologically aligned, differently phrased: *"Pausa temporal de apps de riders durante tormentas"* [Suspend delivery apps during extreme weather] vs *"Bajada del tiempo requerido para la indemnización de transportes"* [Reduce transport worker compensation qualifying period] (8D_sim = 0.990, text_sim = 0.554). Almost no shared vocabulary; the same underlying ideological stance on worker protection and socialised risk. An 8D-only filter would miss that these are distinct proposals worth keeping separate.

The joint filter works: requiring *both* text $\geq$ 0.85 AND 8D $\geq$ 0.85 eliminates both failure modes simultaneously. The threshold is set at 0.85 rather than a lower value: for `paraphrase-multilingual-mpnet-base-v2` on this Spanish political corpus, 21.4% of all nearest-neighbor pairs already exceed 0.70 cosine similarity, making that threshold uninformative for near-duplicate detection. At 0.85, only 0.7% of pairs qualify — these are proposals that genuinely say the same thing and point in the same ideological direction, suitable for collapsing into a single canonical entry without human review. Worked examples of all three cases — text-similar but ideologically opposite, ideologically aligned but differently phrased, and a confirmed near-duplicate cluster — are in Section 7.7.

H3 — Linguistic encoding of ideological axes: All 8 axes have strong vocabulary signal (Spearman ρ range 0.43–0.55, all $p < 0.0001$, N 2,300–2,440 items per axis). This is a stronger result than anticipated from the VIF structure: `technocracy_populism` ($\rho = 0.552$) and `individual_socialized_risk` ($\rho = 0.533$) are the most linguistically detectable axes, while `progressivism_preservation` ($\rho = 0.431$) is the least — but still comfortably above the 0.30 threshold for detectable signal. The implication: for this Spanish political corpus, all 8 axes manifest in surface vocabulary. The 8D framework is not capturing dimensions that text alone is blind to; rather, it is providing a *structured, consistent* ideological interpretation of the same signal that embeddings capture noisily and without semantic commitment. This reframes the value proposition: the scoring pipeline adds structured repeatability, not unique ideological information absent from the text.

H4 — Proposal deduplication clusters: the combined-signal filter (text $\geq$ 0.85 AND 8D $\geq$ 0.85) identified 267 clusters across 53 proposals in the top-10, representing 9,991 aggregate user votes. The threshold was set at 0.85 rather than the conventional 0.70: at 0.70, the `paraphrase-multilingual-mpnet-base-v2` model assigns $\geq$0.70 cosine similarity to 21.4% of all nearest-neighbor pairs for this Spanish political corpus, making the threshold uninformative — union-find chaining at that level collapses unrelated proposals into a single mega-cluster. At 0.85, only 0.7% of pairs qualify, yielding semantically coherent near-duplicate groups. The top 10 clusters by engagement cover: increasing medical staffing (1,809 voters, 10 proposals), state re-industrialisation (1,784 voters, 9 proposals), banning private education (1,430 voters, 13 proposals), and eliminating the self-employed social security flat

rate (1,177 voters, 3 proposals).

4.6 Dimensionality and Product Design

How much of the 8D ideological signal can be captured with fewer dimensions?

Principal Component Analysis on 2,007 valid proposals (items with complete 8-axis scores) shows:

Table 4: PCA explained variance. N=2,007 valid proposals with complete 8-axis scores. 3 components explain 86.2% of total variance.

Component	Explained Variance	Cumulative
PC1	56.2%	56.2%
PC2	16.1%	72.3%
PC3	13.9%	86.2%

PC1 is dominated by `market_collective_allocation` (+0.498) vs `aequitas_libertas` (−0.493), the classic economic left-right axis. PC2 is dominated by `progressivism_preservation` (+0.674) and `universalism_particularism` (+0.565), a cultural axis. The 8D space compresses into a recognisable structure, but meaningful signal remains in the remaining components (Figure 5).

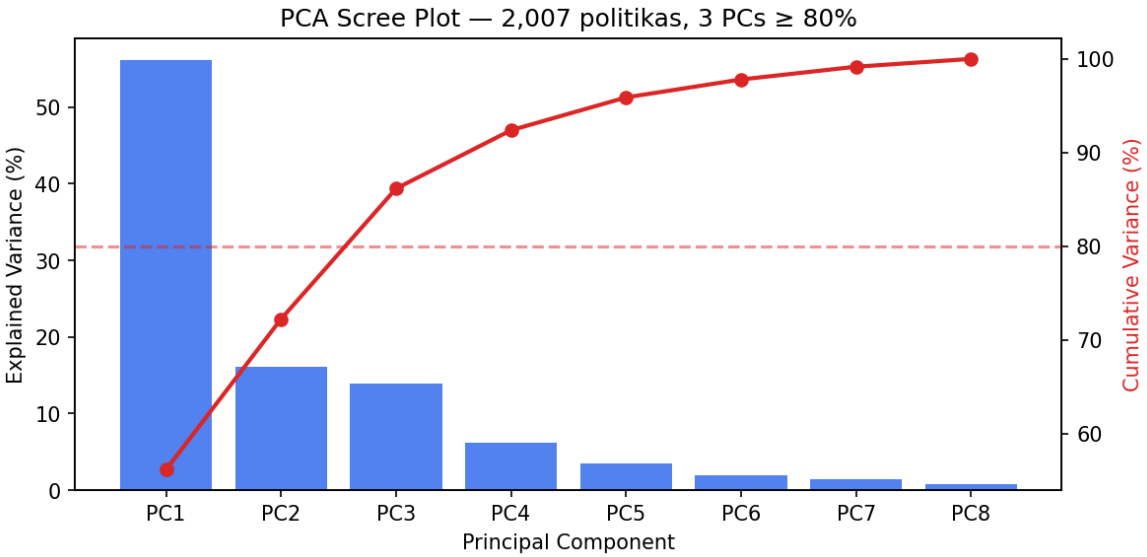

Figure 5: PCA scree plot showing per-component explained variance and cumulative retention. Three components explain 86.2% of total variance; the elbow at PC3 motivates the 3-component summary used in the text.

Axis collinearity and the equity cluster: the economic cluster (`aequitas_libertas`, `market_collective_allocation`, `individual_socialized_risk`, `inequality_acceptance_correction`) is highly redundant; the strongest pair correlation is r = −0.92 between `aequitas_libertas` and `market_collective_allocation`. In retrospect, this was foreseeable: both axes describe the same underlying economic stance from opposite prompt framings. One asks whether resources should be allocated by markets or collectively; the other asks whether individual freedom or equity should prevail. These are the same question with different labels. `inequality_acceptance_correction` (r = −0.85 with aequitas) is

also largely redundant, though it carries a distinct normative connotation (framing inequality as structural and correctable rather than as a neutral economic preference), which may matter for corpus-level interpretation even if the scores collapse empirically. `individual_socialized_risk` (r = −0.80 with aequitas) was the most unexpected redundancy: we expected risk-pooling preferences to be more independent of the equity-freedom dimension.

The contrast with the authority dimension is instructive. The original design intent was to expand symmetrically — three axes elaborating the equity pole, three elaborating the authority pole. The authority cluster held up: `imperium_anarkhia` shows low correlation with the orthogonal axes (r = 0.06–0.33), and `technocracy_populism` (VIF = 1.38) and `universalism_particularism` (VIF = 1.29) carry genuinely independent signal that the economic cluster cannot provide.

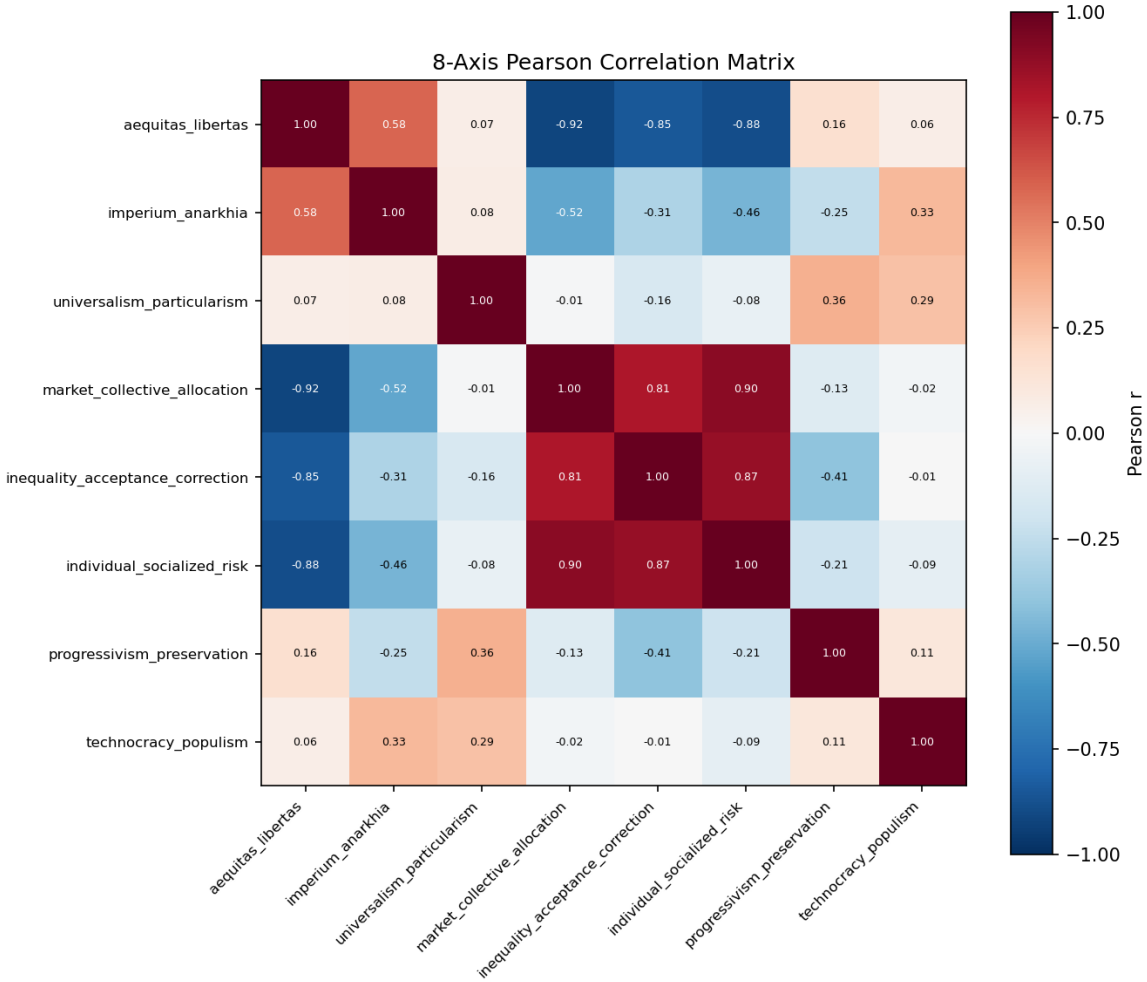

Figure 6: Pairwise Pearson correlation matrix across the 8 ideological axes. The economic cluster (top-left block) shows strong negative correlations (r = −0.80 to −0.92), while the authority-cluster axes are nearly orthogonal to the rest of the space.

Toward a leaner framework: this suggests a consolidation from 8 to 6 axes is warranted. `market_collective_allocation` is the clearest candidate for removal — it is near-semantically identical to `aequitas_libertas` and adds noise rather than signal. `inequality_acceptance_correction` (r = −0.85) is also redundant enough to remove: its normative framing (is inequality a structural injustice or a natural outcome?) sounds

distinct, but in practice scores collapse with the equity axis on this corpus, and the connotation is negative enough to introduce framing bias. `individual_socialized_risk`, by contrast, is a stronger case to retain despite its correlation — the question of who absorbs economic risk (the individual or society collectively) is conceptually separable from the equity-freedom dimension, and on a more ideologically diverse corpus it may decouple. Three realistic paths forward are: (a) drop `market_collective_allocation` and `inequality_acceptance_correction`, arriving at a 6D space that keeps `aequitas_libertas`, `individual_socialized_risk`, and the full authority cluster; (b) maintain the current 2D user-facing product pair (`aequitas_libertas` + `imperium_anarkhia`) for civic interpretability, but revisit whether `imperium_anarkhia` should be replaced by `progressivism_preservation` — the R^2 leaderboard shows the aequitas + progressivism pair is the mathematical optimum ($R^2 = 0.626$ vs 0.595), and progressivism maps to cultural debates that users recognise as well as the authority axis does; or (c) keep the full 8-axis scoring backend but invest in identifying better expansion axes for the equity pole — ones that genuinely add independent signal rather than rephrasing the same economic stance, as the authority expansion (`technocracy_populism`, `universalism_particularism`) successfully did. Which path is best depends on corpus diversity and downstream use case; this paper does not resolve it but makes the trade-off explicit.

Corpus composition hypothesis (imperium aequitas r = +0.58): the moderate correlation between `imperium_anarkhia` and `aequitas_libertas` warrants a corpus-composition interpretation rather than a design one. Category centroids show that 10 of 12 policy categories lean redistributive (negative aequitas mean) and all 12 lean statist (negative imperium mean): Security & Defence is the least redistributive (aequitas = −4.6, near zero) but remains strongly statist (imperium = −45.4). Spain's political discourse, and the Politikea user base, skews left-statist. Proposals that favour redistribution tend also to favour state authority not because these axes are conceptually linked, but because the submitted proposal distribution is not ideologically balanced. A more diverse corpus would likely reduce this correlation. In contrast, `technocracy_populism` (VIF = 1.38) and `universalism_particularism` (VIF = 1.29) are nearly orthogonal to the rest of the space and carry signal the economic cluster cannot provide (see Appendix C, Table 6).

The compression leaderboard: to quantify the cost of dimensionality reduction, we evaluated all 36 possible low-dimensional projections: 8 single-axis (k=1) and 28 two-axis (k=2) subsets, ranked by how much of the full 8D variance structure they reconstruct (mean R^2 via Ridge regression). This comparison makes explicit what is lost when moving from the current standard (a single left-right axis) to a 2-axis framework, and from there to the full 8D space.

Table 5: Compression leaderboard showing selected entries from all 36 possible 1-axis and 2-axis projections, ranked by reconstruction R^2 (Ridge regression predicting all 8 axes, N=2,007 valid proposals). Ranks are non-consecutive; `[...]` marks omitted entries. Full table in Appendix I.

Rank	Projection	k	Mean R^2
1	aequitas_libertas + progressivism_preservation	2	0.626
2	market_collective_allocation + progressivism_preservation	2	0.616

Rank	Projection	k	Mean R^2
3	aequitas_libertas + universalism_particularism	2	0.615
[...]			
5	aequitas_libertas + technocracy_populism	2	0.608
[...]			
11	**aequitas_libertas + imperium_anarkhia** ← Politikea product pair	**2**	**0.595**
[...]			
22	aequitas_libertas + market_collective_allocation	2	0.488
23	**aequitas_libertas** ← best single axis (left-right proxy)	**1**	**0.456**
[...]			
32	universalism_particularism + technocracy_populism	2	0.285
[...]			
36	technocracy_populism	1	0.147

The best single-axis projection — `aequitas_libertas`, the closest analogue to a conventional left-right measure — ranks 23rd of 36 with $R^2 = 0.456$. The Politikea product pair ranks 11th with $R^2 = 0.595$, a 14 percentage-point improvement. Notably, the ranking is not a clean separation: the best single-axis entry outperforms 5 of the 28 two-axis pairs — specifically, those pairing two authority-cluster axes that together miss the dominant economic dimension entirely. This confirms that a *well-chosen* pair matters more than simply having two axes; pairing two low-VIF axes (e.g., universalism + technocracy, rank 32, $R^2 = 0.285$) is worse than using a single economic axis alone.

Why Politikea uses this pair: the product pair was chosen for civic interpretability, not mathematical optimum. The combination of an equity-vs-freedom axis and an authority-vs-decentralisation axis maps to political debates citizens actually recognise and discuss. The mathematically optimal pair (aequitas + progressivism) is harder to explain to a voter who wants to understand *why* two proposals differ.

This is an open question, not a settled answer: is there a 2-axis pair that is both informationally efficient *and* civically meaningful? The data suggests the cost of the Politikea choice relative to the mathematical optimum is approximately 3.1 percentage points of reconstruction fidelity (R^2 0.595 vs 0.626 at rank 1), but the gain over a single left-right axis is 14 percentage points (R^2 0.456 → 0.595). The full 36-entry leaderboard is in Appendix G; the per-axis breakdown is visualised in Figure 7. We do not know if a better pair exists. We are sharing this analysis so others can propose alternatives using the toolkit.

Why 8 axes are still better than 2: the compression leaderboard quantifies the fidelity cost at each dimensionality level. A single left-right axis retains $R^2 = 0.456$ of the 8D structure. The Politikea 2-axis product pair reaches $R^2 = 0.595$ (+14 pp). The full 8D representation, by definition, retains 100% of the signal. Each additional axis beyond the product pair contributes incremental but non-trivial coverage of the remaining 40.5 percent-

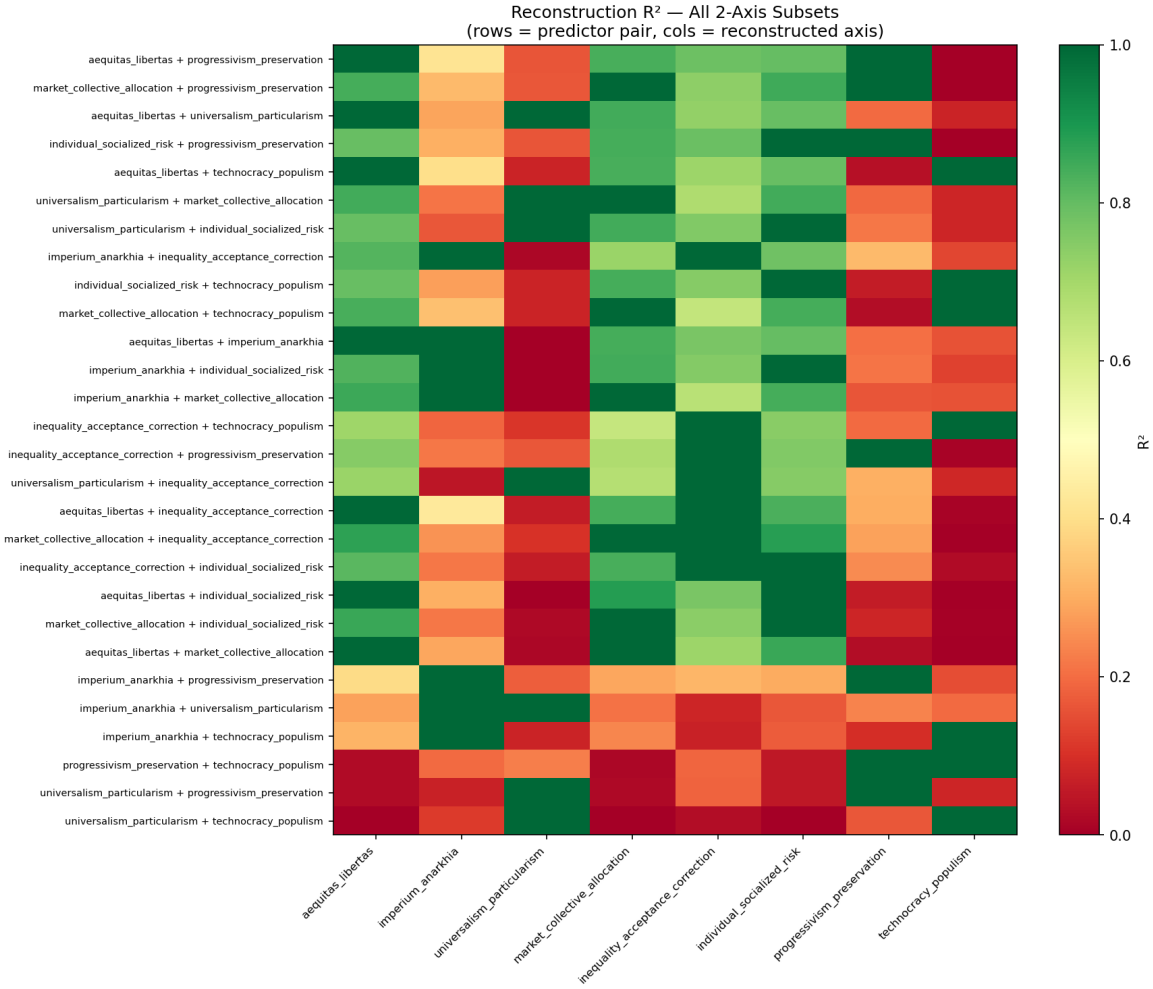

Figure 7: R^2 heatmap for all 28 possible 2-axis pairs (rows) predicting all 8 axes (columns) via Ridge regression. The Politikea product pair (aequitas + imperium) is row 11, with mean $R^2 = 0.595$ vs. 0.626 for the mathematical optimum.

age points; axes like `technocracy_populism` and `universalism_particularism`, which are nearly orthogonal to the economic cluster (VIF 1.38 and 1.29), carry ideological dimensions that no 2-axis projection can recover. This justifies maintaining the full 8-axis backend even when displaying a 2D map to users.

4.7 Platform Behaviour: No Filter Bubble Detected

Politikea's recommendation engine could, in principle, push users toward ideologically narrower proposals over time — a filter bubble effect. We test this directly.

Method: for a cohort of 1,663 users (17.1% of total) with $\geq$100 interactions — the subset with sufficient depth to test the hypothesis — we compute a rolling 8D position vector per interaction bucket (mean of raw axis scores of proposals voted on, over a 10-vote window). Rolling variance measures how much the proposals that users are exposed to scatter across the ideological space. Decreasing variance at the cohort level would indicate gravitation toward a narrower ideological neighbourhood.

Result: variance oscillates between 33.9–42.7 across 0–1,400 interactions with no monotonic trend (Figure 8). Levene's test finds zero statistically significant variance reduction across all 1,663 users (Levene $p < 0.05$ rate = 0.0%).

For this cohort, the ideological diversity of proposals served stays constant from interaction 20 to interaction 1,400. We set out to verify that Politikea does not create filter bubbles for its most engaged users. This is not a guaranteed property of recommendation systems; it has to be measured. We measured it. We will keep measuring it.

Scope caveat: the remaining 82.9% of users have too few votes to test the hypothesis. Sustained engagement is a prerequisite for ideological convergence to become measurable; whether lighter users experience a different pattern is an open question.

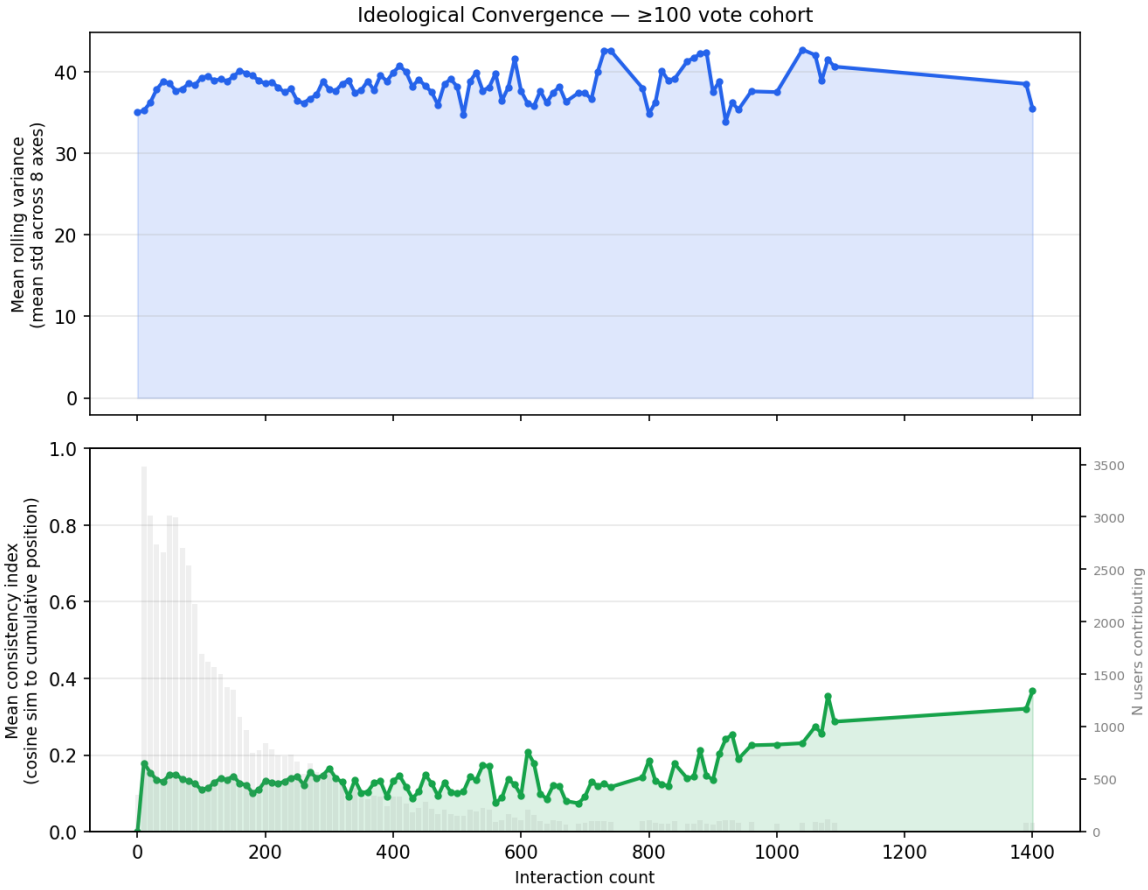

Figure 8: Rolling ideological variance (top panel) and consistency index (bottom panel) across interaction count for the ≥100-vote cohort. Flat variance with no monotonic trend confirms no convergence toward a narrower ideological space across 1,663 users. Bins with <70 users excluded for stability.

5 Limitations and Future Work

No human annotator ground truth: ICC(2,1) is a within-model reliability measure. It tells us GPT-5.2 is consistent with itself. It does not tell us whether GPT-5.2's interpretation of the axes matches human expert consensus. Phase 2 will add a human validation layer.

The 18% reliability tail is uncharacterized: 535 proposals (18.0%) fail at least one reliability threshold and are excluded from downstream analyses. The current framing, "ambiguous proposals that resist clean ideological categorisation," is a hypothesis, not a finding. We have not audited what these proposals actually look like, whether they cluster in particular policy categories, whether they share linguistic features, or whether their failure reflects axis ambiguity in the proposal text versus axis definition gaps in the framework. Two competing explanations have meaningfully different implications: if the proposals are genuinely ideologically ambiguous (e.g., non-partisan administrative measures), exclusion is correct. If the proposals expose axes that are underspecified, lacking enough semantic clarity to be consistently applied, then the right response is axis refinement, not exclusion. Proceeding without this distinction is a real limitation. Phase 2 will characterise the tail before deciding how to treat it.

Spanish-only, single platform: all results are from Spanish political proposals on a single platform with a particular user base. The axis structure, category fingerprints, and convergence result may not generalize to other languages or civic contexts.

Convergence analysis covers only the $\geq$100 vote cohort: 17.1% of users. The result is meaningful for the highly engaged cohort but cannot be extended to the full user base.

The triangulation gate as a diagnostic, not just a rejection: the gate distinguishes correctable from uncorrectable biases, and this distinction is actionable. Gemini's failure is a systematic positive offset — structurally the simplest kind of bias to correct via per-axis mean normalization. After offset correction, the adjusted labels may pass the gate; the failure mode itself tells the practitioner what calibration to attempt. Llama's failure is qualitatively different: coarse quantization and high run-to-run instability are not correctable by simple offset adjustment — they reflect a model that cannot reliably perform the task. The gate's per-axis metrics (directional agreement + MAE) make this distinction visible in the report.

When cross-model audit is unavailable: when a second model is not accessible (cost, API access, latency constraints), the within-model reliability pipeline (ICC filtering, sign agreement thresholds, confidence gating) catches instability of the kind llama exhibited. Systematic biases like Gemini's uniform positive shift could also surface through inspection of score distributions, but cross-model comparison provides a more structured way to characterize them. Together, the two layers form the full validation protocol. We use GPT-5.2 as the primary model throughout this paper because it passes all within-model reliability thresholds; the cross-model audit characterizes the biases each model carries, and frames the calibration work scoped for Phase 2.

Phase 2 roadmap:

- Human annotation layer: expert-labeled proposals to validate GPT-5.2 axis interpretations against human consensus.
- Reliability tail audit: characterise the 535 excluded proposals by category distribu-

tion and linguistic patterns to determine whether exclusion is justified or masks axis definition gaps.

- Axis consolidation: evaluate dropping `market_collective_allocation` and `inequality_acceptance_correction` for a 6-axis framework, or promoting `progressivism_preservation` as a third product axis. Validate on a more ideologically diverse corpus.
- Axis refinement via few-shot calibration: test whether per-axis few-shot examples with explicit polarity anchors reduce instability on axes with elevated tail rates.
- Cross-category consistency: measure whether aggregate ideological patterns hold across policy areas, or shift depending on topic.
- Multilingual expansion: apply the methodology to proposals in other languages.
- Stronger audit models: rerun triangulation with Claude as the primary audit candidate, given its distinct training lineage from the GPT family — maximizing the diagnostic value of cross-model comparison. GPT-4o remains a secondary option for completeness.
- Cross-model alignment: calibration-targeted few-shot examples (not illustrative ones) as the mechanism for aligning heterogeneous models. Section 4.3 shows naive few-shot hurts; the open question is whether targeted examples close the gap.
- Drift analysis: matched before/after proposal content analysis to validate ideological drift detection.

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6 Conclusion

This paper presents three findings about AI-based ideological scoring of political proposals.

First, within-model reliability is strong. GPT-5.2 applies the 8-axis framework with ICC ≥ 0.88 across all axes. 82.0% of 2,977 Spanish proposals pass all reliability thresholds. The framework produces internally consistent labels, stable enough to build on, with absolute calibration against human annotators scoped for Phase 2.

Second, cross-model triangulation reveals that each model carries structurally distinct calibration biases. Gemini 2.5 Flash is internally consistent but systematically biased toward positive scores across all axes: a response-range/acquiescence bias that is correctable via per-axis offset normalization. llama3.1:8b is internally inconsistent due to task overload and limited continuous-scale comprehension, a failure mode that is not correctable by simple adjustment. Both fail the same triangulation gate, but for fundamentally different reasons, and these distinct failure signatures are themselves a useful diagnostic. Absolute calibration of GPT-5.2 against human annotators is scoped for Phase 2.

Third, the 8D framework provides structured, repeatable ideological interpretation of signal that text embeddings capture noisily. It does not uncover dimensions invisible to language — all 8 axes have detectable vocabulary signal ($\rho = 0.43$–0.55) — but it imposes consistent semantic commitment where embeddings offer only geometric proximity. Two

proposals with near-identical wording can sit at opposite ends of the ideological map. The joint filter (text AND 8D similarity) yields a reliable near-duplicate detector. Category-level patterns emerge that are invisible to a left-right slider: security proposals that are simultaneously the most statist and near-neutral on redistribution; rights proposals that cluster universal framing with active reform orientation, directly contradicting how rights discourse is often characterised. The framework also opens a line of inquiry we have not yet closed: are aggregate ideological patterns consistent across different policy areas, or do positions on an axis shift depending on the topic? That question requires more interaction data and is on the Phase 2 roadmap.

The axis set is a starting point, not a final answer. The product pair is a design choice, not a mathematical optimum. The toolkit is the invitation: run this on your proposals, find the patterns we missed, and tell us what we got wrong.

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7 Appendices

7.1 Appendix A — Labeling Prompt (label_8axis_v1.txt)

```
You are labeling a political item across 8 independent ideological axes
on a -100..+100 scale.

Return STRICT JSON only, matching this schema (no markdown, no extra keys):
{
  "axis_aequitas_libertas": <number -100..100>,
  "axis_imperium_anarkhia": <number -100..100>,
  ... (6 more axis scores),
  "conf_axis_aequitas_libertas": <number 0..1>,
  "conf_axis_imperium_anarkhia": <number 0..1>,
  ... (6 more per-axis confidences),
  "global_confidence": <number 0..1>,
  "flags": [<string>, ...],
  "rationale_spans": [{"axis": "<name>", "start": <int>, "end": <int>}, ...]
}

Principles:
- Axes are independent, continuous, bidirectional.
- A score reflects orientation, NOT moral value.
- If unclear: output near-0 and LOW confidence (do not force).

Score interpretation anchors:
- 0 = balanced / mixed
- +/-20 = weak preference
- +/-50 = strong preference
- +/-80..+/-100 = dominant orientation

Axes (short):
- axis_aequitas_libertas:
    -100 equity/redistribution, +100 individual freedom/limited redistribution
- axis_imperium_anarkhia:
    -100 centralized authority, +100 decentralization/self-organization
```

```
- axis_universalism_particularism:
    -100 universal rules/rights, +100 group/context-specific policies
- axis_market_collective_allocation:
    -100 market allocation, +100 collective/democratic allocation
- axis_inequality_acceptance_correction:
    -100 inequality acceptable/natural, +100 inequality must be corrected
- axis_individual_socialized_risk:
    -100 individual bears risk, +100 risk pooled/socialized
- axis_progressivism_preservation:
    -100 change/reform, +100 tradition/continuity
- axis_technocracy_populism:
    -100 expert-led governance, +100 popular-will primacy

rationale_spans:
- 0-4 spans total.
- Offsets refer to the provided TEXT (character offsets).
- Keep spans short and directly evidential.

TEXT:
{{TEXT}}
```

7.2 Appendix B — Threshold Sensitivity Grid

Results are robust across a range of reliability threshold configurations. The highest valid rate achievable with looser thresholds is 87.9% (global_conf = 0.45, axis_conf = 0.35). The configuration used in this paper (global_conf = 0.5, axis_conf = 0.4) is conservative. Full grid available in `outputs/analysis/reports/cleaning_threshold_sensitivity.csv`.

7.3 Appendix C — Variance Inflation Factors

VIF measures how much of an axis's variance is explainable by the other 7 axes. VIF > 5 indicates moderate collinearity; VIF > 10 is severe.

Table 6: Variance Inflation Factors. VIF > 5 = moderate collinearity; VIF > 10 = severe. The top 4 axes form a collinear economic cluster. `technocracy_populism` and `universalism_particularism` are nearly orthogonal to all others.

Axis	VIF
aequitas_libertas	10.44
market_collective_allocation	9.03
individual_socialized_risk	8.20
inequality_acceptance_correction	6.88
imperium_anarkhia	2.50
progressivism_preservation	1.84
technocracy_populism	1.38
universalism_particularism	1.29

7.4 Appendix D — K-Means Clustering

K-means clustering of valid proposals in 8D space. Silhouette = 0.272 at k=5 — weak-to-moderate cluster separation. This is consistent with continuous ideological variation rather than discrete ideological types. Not used as a headline result.

Table 7: K-means silhouette scores. Values below 0.30 indicate weak cluster structure, consistent with continuous rather than discrete ideological variation.

k	Silhouette Score
5	0.272
6	0.262
7	0.259
8	0.238

7.5 Appendix E — Category Names (Spanish Originals)

All category labels in this report use English translations. The original Spanish names as used in the Politikea platform are listed below for reference.

English	Spanish (original)
Culture & Civic Education	Cultura y Educación Cívica
Economy & Taxation	Economía e Impuestos
Identity & Social Cohesion	Identidad y Cohesión Social
Infrastructure & Housing	Infraestructura y Vivienda
Institutions & Government	Instituciones y Gobierno
International Relations	Relaciones Internacionales
Labour & Job Market	Trabajo y Mercado Laboral
Other	Otros
Public Services	Servicios Públicos
Rights & Freedoms	Derechos y Libertades
Science, Energy & Environment	Ciencia, Energía y Medioambiente
Security & Defence	Seguridad y Defensa

7.6 Appendix F — Category Centroid Table

Full mean axis scores per policy category. Axis names abbreviated; full names in Table 1. Negative scores lean left/collective/statist; positive scores lean right/individual/decentralised. Spanish originals in Appendix E.

Table 8: Mean axis scores per policy category (top 8 by size). The 4 smallest categories (Culture & Civic Education, Infrastructure & Housing, International Relations, Science/Energy/Environment) are omitted from this table but present in the full dataset. Axis names abbreviated; full names in Table 1. Negative scores lean left/collective/statist; positive scores lean right/individual/decentralised. Spanish originals in Appendix E.

Category	**N**	**aequitas**	**imperium**	**univers.**	**market**	**inequal.**	**indiv.risk**	**progress.**	**technocr.**
Institutions & Government	499	−5.1	−7.1	−5.0	+19.9	+17.7	+11.9	−28.2	+15.2
Economy & Taxation	412	−0.7	−11.4	+0.2	+3.8	+15.3	+6.4	−27.2	+3.3
Public Services	343	−24.2	−24.5	−14.0	+34.7	+36.3	+34.5	−34.9	−5.2
Security & Defence	216	−4.6	−45.4	+10.3	+22.5	−5.5	+11.9	+14.3	+11.6
Rights & Freedoms	200	−5.2	−17.5	−16.6	+19.2	+20.7	+13.1	−22.9	+7.7
Other	162	−17.3	−34.4	−5.4	+24.1	+16.0	+13.4	−4.2	+8.1
Labour & Job Market	162	−23.7	−15.3	+0.5	+28.2	+37.8	+27.0	−37.6	+6.9
Identity & Social Cohesion	80	−15.7	−26.3	+35.3	+32.2	+9.5	+15.2	+16.7	+27.9

7.7 Appendix G — Joint-Signal Deduplication: Worked Examples

The three examples below ground the abstract similarity scores in concrete proposal text. Each illustrates one failure mode of single-signal deduplication and one success case for the joint filter. Original Spanish text is shown with a brief English gloss in brackets.

7.7.1 Example 1 — Noise Mode 1: Text similar, ideologically opposite (text sim = 0.793, 8D sim = −0.708)

A text-only deduplicator would merge these two proposals — they discuss the same policy area with nearly identical vocabulary (*educación*, *colegios*, *sistema educativo*). The 8D score correctly separates them: one advocates abolishing state-subsidised private schools in favour of universal public education; the other calls for deregulating the entire school system. Same words, opposite stances.

> **Proposal A** — *"Prohibición de la educación concertada"* [Ban on state-subsidised private schools] "Se debe apostar por un modelo de educación pública de calidad donde todo el mundo tenga acceso a los mismos recursos y oportunidades." [We should commit to a quality public education model where everyone has access to the same resources and opportunities.]

> **Proposal B** — *"Desregulación del sistema educativo"* [Deregulation of the education system] "Desregular todo lo relacionado con los colegios, institutos… permitiendo que cada uno pueda crear su propio modelo educativo." [Deregulate everything related to schools and institutes… allowing each school to create its own educational model.]

	text sim	8D sim
Would a text-only filter merge these?	0.793 — **yes**	—
Would a joint filter merge these?	—	−0.708 — **no**

7.7.2 Example 2 — Noise Mode 2: Ideologically aligned, differently phrased (8D sim = 0.990, text sim = 0.554)

An 8D-only filter would flag these as near-duplicates — they share the same ideological stance (worker protection, socialised risk, state intervention in labour markets) to a remarkable degree. But the proposals are substantively distinct: one targets gig-economy apps during extreme weather; the other addresses transport worker compensation timelines. A joint filter correctly keeps them separate.

> **Proposal A** — *"Pausa temporal de apps de riders durante tormentas/olas de frío/calor"* [Temporary suspension of delivery apps during extreme weather] "Apps como Uber Eats/Glovo deberían pausar sus servicios durante tormentas

o olas de calor/frío, para proteger la seguridad de los trabajadores." [Apps like Uber Eats/Glovo should suspend their services during storms or heatwaves/cold snaps, to protect worker safety.]

Proposal B — *"Bajada del tiempo requerido para la indemnización de transportes"* [Reduce the qualifying period for transport worker compensation] "Conforme han ido pasando los años, el tiempo requerido para recibir indemnización… debería reducirse de forma proporcional." [As years have passed, the time required to receive compensation… should be reduced proportionally.]

	8D sim	text sim
Would an 8D-only filter merge these?	0.990 — **yes**	—
Would a joint filter merge these?	—	0.554 — **no**

7.7.3 Example 3 — Joint filter success: Near-duplicate cluster (text sim = 0.932, 8D sim = 0.991)

This pair is the anchor for Cluster 3 in the H4 deduplication results — one of 13 proposals in the corpus that are both textually and ideologically near-identical. All 13 advocate banning private education and strengthening the public system; they represent the same civic demand submitted independently by different users. The joint filter correctly identifies them as candidates for aggregation into a single canonical entry.

Proposal A — *"Prohibición de la Educación Privada"* [Ban on private education] "Prohibir la educación privada y fomentar una enseñanza pública de calidad, polivalente, actualizada y gratuita." [Ban private education and promote quality, versatile, up-to-date, free public schooling.]

Proposal B — *"Prohibición de la educación concertada"* [Ban on state-subsidised private schools] "Se debe apostar por un modelo de educación pública de calidad donde todo el mundo tenga acceso a los mismos recursos." [We should commit to a quality public education model where everyone has access to the same resources.]

	text sim	8D sim
Would a joint filter merge these?	0.932	0.991 — **yes, correctly**

These 13 proposals collectively represent 1,430 user votes on what is effectively a single policy position. Surfacing them as one aggregated entry rather than 13 separate items changes the apparent salience ranking of this demand in the corpus.

7.8 Appendix H — Prompt Ablation Variant Descriptions

Brief descriptions of the 6 prompt variants tested in Section 4.3. All variants score the same 8 ideological axes on a −100 to +100 scale and return JSON output.

Table 9: Variant descriptions for the llama3.1:8b prompt ablation (Section 4.3).

Variant	Description
A: Minimal English baseline	Bare-minimum English prompt: axis names with one-line descriptions and the score range. No anchors, no examples, no reasoning structure. Establishes the floor.
B: GPT-parity English	Full English prompt matching the GPT-5.2 specification in Appendix A — same axis definitions, score anchors (0 = balanced, ±20 = weak, ±50 = strong, ±80–±100 = dominant), and rationale span instructions. Tests whether the richer prompt alone closes the gap when language is held constant.
C: Chain-of-thought (English)	GPT-parity prompt extended with an explicit reasoning step: the model is instructed to produce a short rationale for each axis before committing to a score. English throughout.
D: Nested reasoning JSON	Structured JSON output schema that includes a `reasoning` field per axis alongside the score. Forces the model to externalise intermediate reasoning as part of the output contract, rather than as free text. English throughout.
E: Few-shot calibration examples	GPT-parity prompt augmented with 2 scored example proposals (one clearly redistributive, one clearly libertarian) with expected scores. Tests whether concrete anchors improve scale usage.
F: Spanish, one axis per call	Prompt and axis definitions translated to Spanish; the 8-axis scoring task decomposed into 8 separate API calls (one axis per call). Reduces task complexity and aligns prompt language with corpus language. **Recommended baseline for Spanish-corpus replication.**

7.9 Appendix I — Full Compression Leaderboard (1D + 2D)

All 36 possible low-dimensional projections (8 single-axis + 28 two-axis), ranked by reconstruction R^2 via Ridge regression predicting all 8 axes. N=2,007 valid proposals.

Table 10: Full 1D+2D compression leaderboard. Bold rows: Politikea product pair (rank 11) and best single-axis projection (rank 23, left-right proxy). The best 1D entry outperforms 5 of the 28 2-axis pairs — specifically those pairing two authority-cluster axes that together miss the dominant economic dimension.

Rank	Projection	k	Mean R^2
1	aequitas_libertas + progressivism_preservation	2	0.626

Rank	Projection	k	Mean R^2
2	market_collective_allocation + progressivism_preservation	2	0.616
3	aequitas_libertas + universalism_particularism	2	0.615
4	individual_socialized_risk + progressivism_preservation	2	0.612
5	aequitas_libertas + technocracy_populism	2	0.608
6	universalism_particularism + market_collective_allocation	2	0.607
7	universalism_particularism + individual_socialized_risk	2	0.607
8	imperium_anarkhia + inequality_acceptance_correction	2	0.601
9	individual_socialized_risk + technocracy_populism	2	0.600
10	market_collective_allocation + technocracy_populism	2	0.596
11	**aequitas_libertas + imperium_anarkhia**	**2**	**0.595**
12	imperium_anarkhia + individual_socialized_risk	2	0.595
13	imperium_anarkhia + market_collective_allocation	2	0.584
14	inequality_acceptance_correction + technocracy_populism	2	0.574
15	inequality_acceptance_correction + progressivism_preservation	2	0.572
16	universalism_particularism + inequality_acceptance_correction	2	0.571
17	aequitas_libertas + inequality_acceptance_correction	2	0.560
18	market_collective_allocation + inequality_acceptance_correction	2	0.549
19	inequality_acceptance_correction + individual_socialized_risk	2	0.525
20	aequitas_libertas + individual_socialized_risk	2	0.501
21	market_collective_allocation + individual_socialized_risk	2	0.490
22	aequitas_libertas + market_collective_allocation	2	0.488
23	**aequitas_libertas**	**1**	**0.456**
24	imperium_anarkhia + progressivism_preservation	2	0.453
25	individual_socialized_risk	1	0.449
26	market_collective_allocation	1	0.444
27	inequality_acceptance_correction	1	0.418

Rank	Projection	k	Mean R^2
28	imperium_anarkhia + universalism_particularism	2	0.396
29	imperium_anarkhia + technocracy_populism	2	0.372
30	progressivism_preservation + technocracy_populism	2	0.339
31	universalism_particularism + progressivism_preservation	2	0.304
32	universalism_particularism + technocracy_populism	2	0.285
33	imperium_anarkhia	1	0.235
34	progressivism_preservation	1	0.191
35	universalism_particularism	1	0.155
36	technocracy_populism	1	0.147

7.10 Appendix J — Supplementary Material

Source code and toolkit: https://github.com/politikea/ideology-mapping

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